Using Restricted-Access ACS Data to Examine Economic and Noneconomic Factors of Interstate Migration By Race and Ethnicity

by

Bryanna Duca*
Starbucks Corporation

Anita Alves Pena
Colorado State University

CES 23-12 **March 2023**

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Abstract

We explore how determinants of internal migration differ between Black non-Hispanics, White non-Hispanics, and Hispanics using micro-level, restricted-use American Community Survey (ACS) data matched to data on attributes of sub-geographies down to the county level. This paper extends the discussion of internal migration in the U.S. by not only observing relationships between economic and noneconomic factors and household-level propensities to migrate, but also how these relationships differ across race and ethnicity within smaller geographies than have been explored in previous literature. We show that when controlling for household and location characteristics, minorities have a lower propensity to migrate than White households and document nuances in the responsiveness of internal migration to individual and locational attributes by racial and ethnic population subgroups.

Keyword*:* migration, race and ethnicity, restricted ACS data

JEL Classification: J15, R23, C5

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[*] **Disclaimer**: Any views expressed are those of the authors and not those of the U.S. Census Bureau. This research was conducted while Duca was a graduate student at Colorado State University. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product (authorization numbers 9261 and 9563).

Acknowledgements: We acknowledge the support of Drs. Jani Little and Philip Pendergast of the Rocky Mountain Federal Statistical Data Research Center and the helpful feedback on earlier drafts provided by Drs. Stephan Weiler, Alex Bernasek, and Joon K. Kim of Colorado State University. A previous version of this research appeared as part of a PhD dissertation at Colorado State University: Duca, Bryanna. Socioeconomics of Modern-Day Migration Within the United States: Determinants and Economic Implications Across Race and Ethnicity. Colorado State University. Libraries, 2022. Print.

1 Introduction

While the propensity to migrate within the U.S. population overall has decreased over the past couple decades, relocation remains necessary at points in the life cycle for some households. Incentives to relocate may relate to financial or cultural opportunities or personal circumstances such as returning to a childhood home, moving to be closer to family, or relocating to get away from a current living situation. Understanding motivations for migration provides valuable information toward analyzing demographic, economic, and social outcomes. Using 2007 to 2015 micro-level restricted-use data published by the American Community Survey (ACS) combined with state- and county-level data, this paper explores relationships between economic and noneconomic location and household characteristics. For the purposes of this research, moves within county are considered residential movement, while movement outside of the original county is considered migration. In both cases, we only consider internal movements within the U.S. leaving the dynamics of international migration to other work. Using Heckman corrected probit, we model characteristics of people and place in a two-step decision problem where households first determine whether or not to migrate and second determine whether to stay within a given location or to move to another. We examine determinants of migration for the representative U.S. sample as a whole

and then separate by Black non-Hispanic, White non-Hispanic, and Hispanic.[1]

Research on the determinants of migration has improved with increased availability of micro data and an evolution of the understanding of motivations behind migration decisions. Beginning as early as Ravenstein (1885), literature on internal migration has sought to isolate understanding of who migrates by identifying individual characteristics such as age, education, income, and marital status, and to where they migrate by examining location-specific characteristics. Sociology literature has furthered the study of internal migration by suggesting how the development of sociological groups and support networks may affect migration patterns, providing reason to expect systemic racial and ethnic gaps in migration within the U.S. (Greenwood 1975). Finally, some research has added noneconomic determinants (often defined in terms of quality of life factors) such as access to public goods or location-specific amenities (Cebula 2005).

While micro-level data have become more readily available for research purposes, public-access data remain limited. An implication of limitations in the public availability of complete micro-level data is that researchers have had to get creative in how they explore internal migration patterns. Pais (2014), for example, used a longitudinal study in order to analyze repeat migration while other papers restricted analysis of internal migration to higher geographical levels such as state-level. With access to restricted-use ACS data, our paper, in contrast, uses a more narrow definition of migration that allows for the identification of nuances in population dynamics. We follow the definition of migration provided by the Population Association of America. Migration is defined as "a relatively permanent change of residence that crosses jurisdictional boundaries (counties in particular), measured in terms of usual residence at a prior point in time, typically 1-5 years earlier" (Greenwood 1997).

Restricted-use ACS data provides unconstrained data such as the elimination of income caps on reported household income and the observation of finer geographic levels. These data also allow findings of internal migration to be further disaggregated by race and ethnicity, in this case for the groups of Blacks, Whites, and Hispanics as previously mentioned. While many household and location characteristics have similar relationships with the propensity to migrate across race and ethnicity, many magnitudes differ across subgroups especially in the presence of controls for

[1]For simplicity, we refer to Black non-Hispanics and White non-Hispanics as Black and White respectively in the text. Due to sample sizes at small geographies, we do not extend to additional racial and ethnic categories.

economic and noneconomic characteristics. These findings are documented in this paper.

2 Literature and Conceptual Background

Both economic and noneconomic literature is used to build the model and hypotheses of determinants of internal migration across race and ethnicity. Categories of literature are used as a means to construct the necessary model but also to analyze results. While historical findings are used toward explaining differences in behaviors across race and ethnicity, sociological literature provides insight into necessary variables that theoretically influence migration. We also use past literature to motivate the use of noneconomic factors that influence migration, such as the strength in social networks, and potential methods to model them.

2.1 Economic Determinants of Internal Migration

Michael J. Greenwood defined the determinants of migration as "the factors that affect migration, including characteristics of both places and of persons and their families" (Greenwood 1997). Some of the most important economic characteristics are household and individual differences that make a household unit or its members more or less likely to migrate. Age (of the household head or others in the family unit) and the propensity to migrate are expected to share a significant, non-linear relationship (Plane 1993). Another important economic determinant is educational attainment (Greenwood 1997). Using a longitudinal study, Pais (2014) used educational attainment rather than an income measure to examine repeat migration. As explained by Patillo-McCoy (2000), class structure also influences mobility. This author recognizes the differences in spatial contexts of the Black and White middle class and examines the importance of race in determining spatial and social outcomes. He argues that "changes in the socio-economic distribution of African Americans can be translated into changes in their spatial distribution" (Patillo-McMoy 2000).

In addition to individual specific characteristics, features of the location itself can either pull or push migrants. Furthermore, some factors can act as an anchor in keeping residents within a particular locale. A motivating location specific factor of migration, for example, is employment. Holzer (1991) compiled 20 years' worth of evidence on the spatial mismatch hypothesis. The most recent evidence suggests that spatial mismatch is relevant for explaining Black and

White employment differences. The trends are relevant when considering mismatch related to continuing decentralization of population and employment in metropolitan areas that changes the characteristics of employment in the U.S. generally and among Blacks in particular.

2.2 Noneconomic Determinants of Internal Migration

The inclusion of noneconomic factors is important in building a complete model of migration. Price-Spatlen (1998) illustrated the changing significance of noneconomic factors in shaping the influences of Black migration. Both Price-Spatlen (1998) and Enchautegui (2016) identify one of these noneconomic factors as ethnogenesis. "Ethnogenesis is the process by which ethnic and racial groups refine a sense of urban place by developing and refining a communal structure and a collective ethos from the interplay between socio-cultural characteristics and American social structure" (Enchautegui 1997). Authors have found that urban areas with strong ethnogenic measures not only attract potential migrants but also serve as an anchor in keeping Blacks in that location. Enchautegui (2016) found that minority groups prefer to live in areas where people of their race or ethnicity are a larger proportion of the population (relative to other locations) as a means toward taking advantage of social and economic support of spatially concentrated ethnic networks. It was through the process of ethnogenesis in the 1930's that significantly shaped both total Black migration and migration of young Blacks, although it was not as much of a driving force in the 1950's. These papers help to support the claim made in Liao (2012) that "individuals who are not White are less likely to move" (Liao 2012). The establishment of strong communities from previous waves of migration of minorities contributes to a decrease in residential mobility.

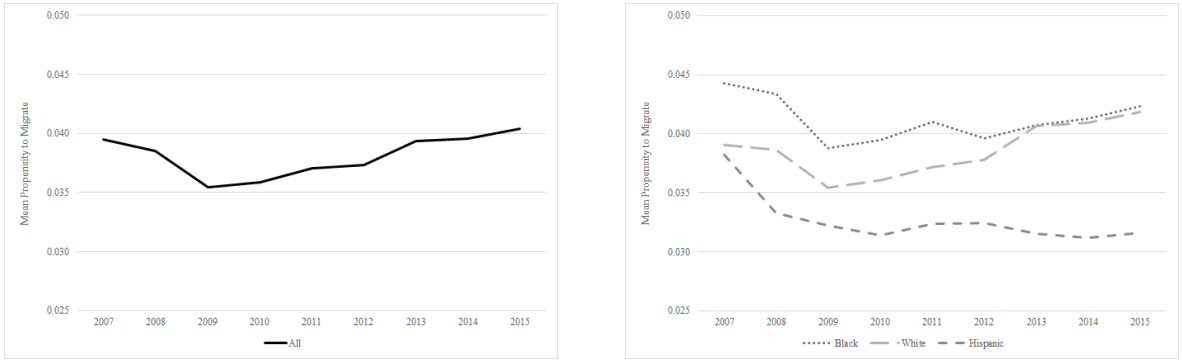
3 Data

We analyze household migration decisions using restricted-use ACS data from 2007 until 2015. In addition to using the ACS for individual and household level characteristics, we match data on state- and county-level characteristics gathered from multiple sources such as the Bureau of Labor Statistics (BLS), the Bureau of Economic Analysis (BEA), and the Census Bureau. The sample analyzed in this paper includes all household heads that self-identify as either Black, White, or Hispanic, who are age 18 or older, and who were neither a full-time student nor active military

within the last year. Eliminating full-time students and active military avoids those households that likely were forced to move for either education purposes or for military relocation.

3.1 Snapshot of Internal Migration in the U.S.

Figure 1 presents time trend diagrams representing the mean migration rate across race and ethnicity. Migration rates were calculated by creating a binary variable that measures whether the household migrated to at least a different county (*Migrate*=1) or stayed in the same residence or moved within county (*Migrate*=0). The definition of this variable follows the definition provided by the Population Association of America noted above.



(a) Entire Sample (b) Across Race & Ethnicity

Figure 1 Mean Propensity to Migrate

The propensity to migrate declined over the starting part of the series presented in Figure 1. Migration rates then increased during the 2008 recession as a response to major economic shocks that contributed to rapid spikes in unemployment, decreases in disposable income, and an increase in housing insecurity. This aligns with previous literature on the decline in internal migration rates noting the exception of recessionary periods and their prospective recovery (Cooke 2013; Monras 2015; Molloy et al. 2011). The decline in internal migration coincides with theories of ethnogenesis along with increased importance of social networks and social capital (Haug 2008). The affinity hypothesis, states that more friends, family, and investment in a community reinforce the tendency not to migrate.

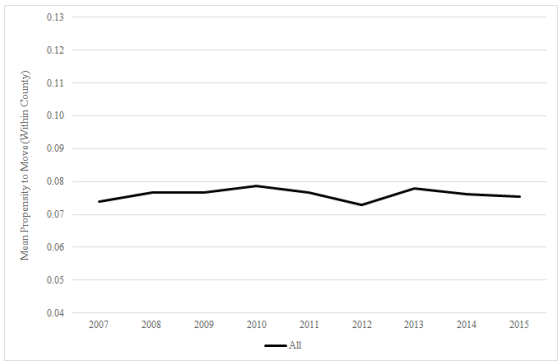
Across race and ethnicity, the second panel of Figure 1 shows that Black household heads had higher rates of country-level migration, with the gap between White and Black household rates closing in 2013. This is different from the claim made in Liao (2012) and in other literature

suggesting that Blacks are less likely to migrate. One reason for this difference is the differing definitions of migration. Another significant difference is that these figures do not control for various other factors that can influence the propensity to migrate. Our definition excludes households that are moving within county, which likely captures residential displacement and may disproportionately impact Black households. This also aligns with ethnogenesis and social network theories that anchor Black households within a particular area. Hispanic household heads have the lowest propensity to migrate by this country-level definition and they saw the smallest increase in migration during the Great Recession. Minority households were disproportionately impacted by the Great Recession. Why changes in Black household county to county movement increased while that for Hispanics was relatively constant also could relate to Hispanic households moving within county more during this time along with labor demand impacts for agricultural and industrial sectors laborers (Villarreal 2014). Theoretically, fewer employment opportunities elsewhere serves as a strong locational anchor, disincentivizing migration.

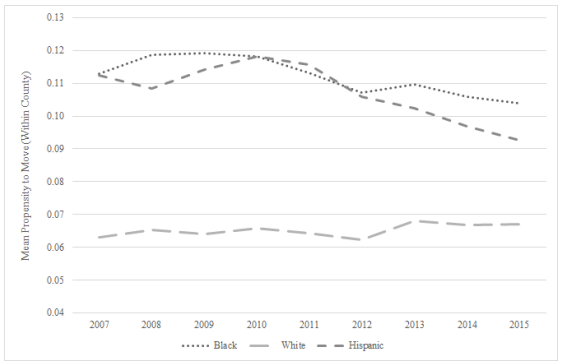
Figure 2 presents time series trends representing the propensity to move where moving refers to households that moved within the same county in the last year. Moving rates were calculated by creating a binary variable equal to 1 if the household moved within the same county in the last year and 0 otherwise. This means that a household that migrated outside of the county have a value of 0. For the entire sample, we see little change in the propensity to move over the 10-year time period. However, we see more variation when disaggregating by race and ethnicity. Hispanic and Black households have a higher propensity to move than White households. This is consistent with literature on residential displacement, gentrification, and housing discrimination. In 2010, Black and Hispanic households were almost twice as likely to move within county than White households.

Figure 3 measures the propensity to participate in interstate migration, moving to a different state, rather than intrastate migration, moving within state but to a different county. The propensity to participate in interstate migration is measured with a binary variable equal to 1 if the household migrated to a different state (given the household migrated) and 0 if the household migrated within state in the last year. For the entire sample, between 42% and 45% households migrated to a different state while the remaining migrated within state.

Strong social networks, family structure, and economic costs of migrating can help to explain



(a) Entire Sample

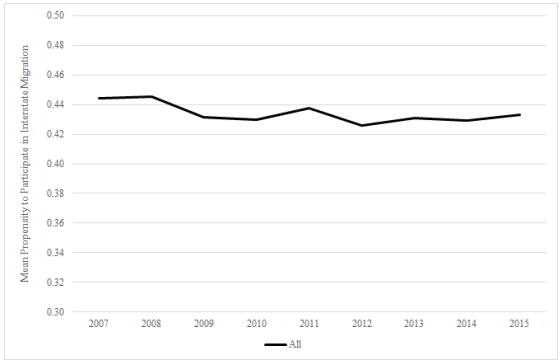


(b) Across Race & Ethnicity

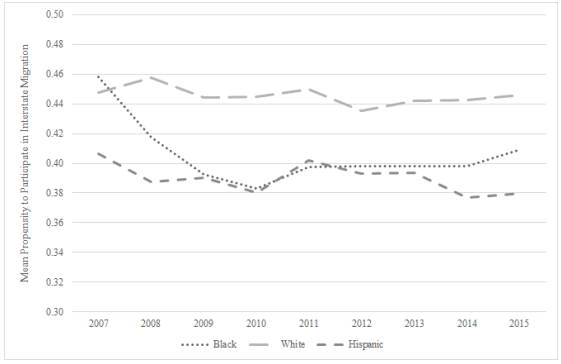
Figure 2 Mean Propensity to Move Within County

the lower propensity to migrate to a different state. When households migrate, White households are more likely to participate in interstate migration than Black and Hispanic households. Black households experienced similar rates of interstate migration in 2007 but saw a drastic decrease during the Great Recession. Minorities were disproportionately impacted by the recession. Without the means or opportunities motivating further migration, it makes sense that these households saw a dip in interstate migration.

The three sets of figures represent the propensity to migrate across counties, to move within counties, and to participate in interstate migration respectively without controlling for various factors that impact residential mobility. By including household and location characteristics in the model, we can examine correlations between race and ethnicity and internal migratory behavior.



(a) Entire Sample



(b) Across Race & Ethnicity

Figure 3 Mean Propensity to Migrate to a Different State

3.2 Individual and Household Determinants of Internal Migration

Since we are interested in how both economic and noneconomic individual and location characteristics motivate inter- and intrastate migration within the U.S., Table 1 summarizes the economic individual characteristics of the household heads included in the analysis provided by the ACS data from 2007 to 2015.

Table 1 List of Household and Individual Characteristics

Economic
Family Size
Number of Children Age 5 and Above
Number of Children Under the Age of 5
Sex (Male = 1)
Age (Years)
Marital Status (Married = 1)
U.S. Born (Yes = 1)
Educational Attainment (measured using a series of binary variables)
Labor Force Participant (Yes = 1)
Total Household Income

Based on the definitions of economic versus noneconomic determinants of migration in Cebula (2005), all individual and household level characteristics are classified as economic determinants. All of the household and individual characteristics are provided by the same restricted-use ACS data that is used to construct the dependent variables of interest. As defined by the ACS, family size refers to the number of family members that are residing in the household head's places of residence permanently. This can include in-laws, nieces and nephews, grandparents, etc. Theoretically, a growing (or shrinking) household could induce migration but may have a larger impact on location and distance of migration. Children in particular can hold strong deterministic power. The number of children, defined as the household head's "own children" [2], is split up into two categories: number of children age 5 and above and the number of children under the age of 5. Ham (2015) finds that the more children within a family, the less likely the family unit is to migrate. While we hypothesize the same, we distinguish between school-age children and infants/toddlers. Families with young children that are not in school are hypothesized to be more mobile than households with school-age children who have their own social networks, making it harder for household to move longer distances. It also is possible that households are more likely to migrate or move to areas with higher ranked schools while their children are still young. The

[2] "Own children" includes biological, step, and adopted children.

summary statistics are provided in Table 2 for all three of these variables but only the two children variables are ultimately included in the model to avoid high correlations with family size.

Age is a continuous numerical variable with a minimum value of 18. The ACS defines total family income as the total pre-tax income earned by the family. Total family income represents the total income earned by all members in the household, therefore total family income of a household of one would equal individual income. Income variables are adjusted for 2015 dollars. A series of education related binary variables also are included. Educational attainment has been broken up into five categories: less than high school, high school, some college, college, and advanced degree. Due to high correlations between income and educational attainment and the possibility of endogeneity, only educational attainment variables are included in the final models. The binary variable that measures the sex of the household head equals 1 if male and 0 if female.[3] Marital status equals 1 if married and 0 otherwise. The additional binary variables included are: "U.S. Born" and "Labor Force Participation" each equaling 1 if the respondent falls into that category and 0 otherwise.

Table 2 summarizes the individual and household characteristics of the sample of household heads disaggregated by race and ethnicity in the final sample using person weights provided by the ACS. As shown in the figures, Black households are more likely to migrate and both Black and Hispanic households are more likely to move within county. Household and individual characteristics generally follow expected patterns. Minority households on average earn less, have a lower propensity to be married, have more children, and are less likely to own their home. Black and Hispanic households heads are more likely to be less educated and less likely to hold an advanced degree. It is important to recognize these differences as they can help to explain differences in the propensity to migrate. By controlling for many of these characteristics, we are better able to separate the relationship between having one of these characteristics on one's residential mobility rather than the racial and ethnic impact itself.

[3]Throughout this paper, we use the terms sex and gender interchangeably. When citing and referring to findings from previous publications, we match their terminology. The ACS data names the self reported variable as sex.

Table 2 Summary Statistics of Economic Household and Individual Characteristics

	All	Black	White	Hispanic
Black	0.1173			
	(0.0001267)			
White	0.7528			
	(0.0001668)			
Hispanic	0.13			
	(0.0001327)			
Migrate	0.03806	0.04117	0.03852	0.03261
	(0.0000755)	(0.0002532)	(0.0000847)	(0.0002118)
Mover	0.07601	0.1121	0.06503	0.1071
	(0.0001056)	(0.0003906)	(0.0001093)	(0.0003624)
Average Personal Income (2015 dollars)	$46,900	$32,770	$51,560	$32,690
	(21.56)	(43.66)	(26.53)	(42.93)
Average Household Income (2015 dollars)	$72,880	$49,170	$79,420	$56,370
	(28.13)	(60.83)	(34.15)	(60.23)
Sex (Male = 1)	0.5098	0.3835	0.5255	0.533
	(0.0001791)	(0.0005711)	(0.0001978)	(0.0005541)
Age	52.14	49.4	53.65	45.8
	(0.005723)	(0.01697)	(0.006466)	(0.01503)
Marital Status (Married = 1)	0.4982	0.2875	0.528	0.5155
	(0.0001791)	(0.0005225)	(0.0001984)	(0.0005561)
Family Size	2.408	2.395	2.275	3.19
	(0.0005397)	(0.001782)	(0.0005408)	(0.0020)
Children Age 5 and Up	0.5952	0.6966	0.5029	1.038
	(0.0003582)	(0.001235)	(0.000358)	(0.001326)
Children Under the Age of 5	0.1279	0.1263	0.1109	0.2277
	(0.0001584)	(0.0005176)	(0.0001624)	(0.0006056)
Homeowner (Yes = 1)	0.684	0.4621	0.7484	0.5116
	(0.0001766)	(0.0005758)	(0.0001852)	(0.0005569)
Labor Force Participant (Yes = 1)	0.6794	0.6857	0.6636	0.7651
	(0.0001625)	(0.000518)	(0.0001833)	(0.0004554)
Less than High School	0.1236	0.1537	0.07972	0.3511
	(0.0001221)	(0.0004035)	(0.0001084)	(0.0005309)
High School	0.2643	0.2854	0.2634	0.2505
	(0.0001588)	(0.0005277)	(0.0001749)	(0.0004881)
Some College	0.225	0.2653	0.2255	0.1858
	(0.0001513)	(0.0005193)	(0.0001678)	(0.0004333)
College	0.2711	0.2163	0.2984	0.162
	(0.0001581)	(0.0004858)	(0.0001812)	(0.0004055)
Advanced Degree or Higher	0.116	0.07935	0.133	0.05061
	(0.0001078)	(0.0003041)	(0.0001285)	(0.0002252)
U.S. Born (Yes = 1)	0.8726	0.9048	0.9484	0.4042
	(0.0001293)	(0.0003554)	(0.0000914)	(0.0005431)
Urban Resident	0.7868	0.898	0.7457	0.9245
	(0.0001407)	(0.0003535)	(0.0001679)	(0.0002974)
N	12,240,000	1,140,000	9,895,000	1,204,000

[a] Standard errors are in parentheses.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.

3.3 Location Determinants of Internal Migration

In addition to examining the explanatory role of individual and household characteristics on migratory behavior, we analyze the motivating power of location characteristics. Previous literature suggests the importance of not only including characteristics of the current place of residence but also characteristics of the previous location. Location characteristics of the origin are used in analyzing the household's decision to migrate and differences of the location characteristics of the origin versus the destination are used in explaining inter- versus intrastate migration. For example, the difference in the Black proportion of the population is equal to the percentage of the population that identifies as Black from the current location minus the percentage of the population that identifies as Black from the previous location. Data on location characteristics was only available as early as 2007, which determines the start year of our analysis. Table 3 lists the different location characteristics, separated by household-specific economic and noneconomic variables.

Table 3 Location Specific Characteristics

Non-Economic
Population Composition (Percentage of population that is Black, White, or Hispanic)
Per Capita Expenditures on Parks and Recreation (2015 dollars)
Economic
Monthly Median Housing Prices (2015 dollars)
Per Capita Income (2015 dollars)
Unemployment Rate
Employment Growth (year over year)
Population

Following the literature, we think of noneconomic variables as quality of life variables. These factors hold little to no economic value from the individual perspective but can increase or decrease utility of households. To proxy quality of life, county level per capita expenditures on parks and recreation[4] is used and is adjusted for inflation using 2015 dollars. In order to test the theory of ethnogensis, population composition variables are included. These variables are calculated by dividing the total population of a particular race or ethnicity provided by the ACS by the total population of that particular county. Monthly median housing costs[5] are at the state-level and

[4]U.S. Census Bureau, Government Finance Database

[5]U.S. Census Bureau, American Community Survey (2007-2015)

include both monthly owner and renter costs. Per capita income[6] is at the county level, providing an additional measurement of the standard of living within a particular location. Employment growth is the year over year change in employment totals provided by the BEA at the county level. The unemployment rate[7] is represents the county-level percentage of the labor force that is unemployed.

4 Empirical Methodology

Household units are envisioned to maximize lifetime well-being by taking into consideration a multitude of factors. Theoretical modeling of household decision-making is presented in Appendix A. The probability of migration then is a function of individual and household characteristics and location characteristics. The first question that must be answered by household m is whether or not they should migrate from location j at time period t. The dependent binary variable, $Migrate_{mjt}$ is defined as:

$$Migrate_{mjt} = 1 \textit{ if household m migrates anywhere}$$

$$Migrate_{mjt} = 0 \textit{ if household m stays within the same residence}$$

The second question is the location of migration, measured by whether or not the household m participates in interstate migration. The second dependent binary variable of interest $Interstate_{mjt}$ is defined as:

$$Interstate_{mjt} = 1 \textit{ if migrant(s) migrates to a different state}$$

$$Interstate_{mjt} = 0 \textit{ if migrant(s) migrates within state}$$

The empirical strategy for analyzing the effect of race and ethnicity on migration is to estimate the probability of migrating conditional on variables that have historically influenced migration, including both individual and household characteristics. The Heckman corrected Probit estima-

[6]Bureau of Economic Analysis, GDP & Personal Income (2007-2015)
[7]U.S. Bureau of Labor Statistics (2007-2015)

tion can be formalized as such:

$$Pr(Migrate_{mjt} = 1) = f(\alpha_1 + X_{jt}^T\beta + Z_{jt}^T\gamma + V_{mt}^T\delta) \quad (1)$$

$$Pr(Interstate_{mjt} = 1) = f(\alpha_2 + W_{ijt}^T\beta + Q_{ijt}^T\gamma + V_{mt}^T\delta) \quad (2)$$

where $Pr(Migrate_{mjt} = 1)$ is the selection group and represents the probability of household m migrating from place j at time period t and $Pr(Interstate_{mjt} = 1)$ represents the probability of household m migrating from state j to state i, provided they are a part of the selection equation. The model assumes the probability of participating in interstate migration is observed only if $Migrate_{mjt} = 1$.

The probability of migrating is a function of a vector of location-specific economic characteristics, X_{jt}, where location is defined in terms of the origin, j. The variables within this vector include monthly median housing costs, per capita income, unemployment rate, and the size of the population. The probability of migrating also is a function of origin specific non-economic factors, Z_{jt}. This includes an ethnogenesis measure calculated as the percentage of the location's population that identifies as Black, White, or Hispanic. It also includes per capita expenditures on parks and recreation. V_{mt} represents a vector of household and individual characteristics consisting of binary race variables, marital status, number of children age 5 and above, number of children under the age of 5, educational attainment, and other demographic characteristics. The probability of participating in interstate migration (rather than intrastate migration) is a function of the same vector of economic and non-economic household and individual characteristics and a vector of non-economic location characteristic differences, Q_{ijt}. Location differences are the mathematical differences between the origin's and the destination's characteristics as previously described.

5 Results

The Heckman Probit model breaks the household's decision to migrate into a two-step process. We discuss these steps in turn in what follows. We examine the marginal effects of the selection portion of the equation, whether the household has migrated within the last year followed by

results across the different racial and ethnic groups separately. Afterwards, we examine how determinants influence interstate versus intrastate migration as the second step of the model's process.

5.1 Determinants of the Selection group: Migrate

Table 4 provides the marginal effects of the first step of the Heckman selection model for the entire sample. A one unit change in the independent variable increases (or decreases) the propensity to migrate by β percentage points, where β is the coefficient of the independent variable. With the inclusion of both time and region fixed effects, male household heads are 0.025 percentage points more likely to migrate than female households, and both Black and Hispanic households are less likely to Migrate than White household when controlling for various other individual and household characteristics, consistent with the findings of previous literature and in contrast to the unconditional statistics presented in the earlier figures. Age has a statistically significant quadratic relationship with the propensity to migrate. Married household heads are 0.22 percentage points less likely to migrate as the household head has to consider the opportunity costs of moving not only themselves but their partner as well.

Another life stage that statistically significantly impacts residential mobility is having children. Households with children are less likely to migrate and are even less mobile in the presence of school-age children over the age of 5. As shown in Table 4, households with school-age children are 0.12 percentage points less likely to migrate while households with children under the age of 5 are 0.035 percentage points less likely. School-age children begin building their social networks as they enter school, increasing the noneconomic costs of migrating. While having more children in general decreases residential mobility, this decreases is magnified when children are school-age.

Household heads with less than a high school degree, some college, college, or an advanced degree are more likely to migrate than are those with only a high school degree. Those that have completed college or an advanced degree are even more likely to migrate. This could be a result of higher educational attainment's impact on income and therefore on the means to migrate or could be a result of employment opportunities and the need to migrate for a particular job.

In addition to individual and household characteristics influencing internal migratory behavior, the characteristics of the current place of residence can anchor or push migration. The unemploy-

ment rate of the county has a statistically significant positive relationship with migration. A 1 percentage point increase in the unemployment rate of the county, increases the likelihood of the household migrating by 0.002 percentage points. This is consistent with previous literature on the relationship between economic shocks and internal migration (Monras 2015). Locations that are negatively impacted by an economic shock, usually characterized by high unemployment, see more out-migration than do locations that are less impacted.

The population of the location also influences households' decisions to migrate. Locations with a higher population experience a lower propensity to migrate. Although racial and ethnic population portions are included in the entire sample, it is more insightful to examine the migratory behaviors for each race and ethnicity separately as the relationships across race and ethnicity are expected to differ. Monthly median housing costs are positively correlated with the propensity to migrate. As it becomes more expensive to live in a particular location, the benefits of migrating increase and households may choose to leave or be forced to migrate due to costs of living. While costs push households to migrate theoretically, an increase in benefits should anchor or pull migrants to a specific place. County-level spending on parks and recreation is negatively correlated with migration. As per capita spending on amenities increases, migration decreases.

Individual, household, and location factors can influence the economic and noneconomic costs and benefits of migrating. Table 5 uses the same model specifications as Table 4 with the exception of the race binary variables as the model instead is estimated for each racial and ethnic group separately. The results include both time and region fixed effects.[8]

Sex, age, marital status, and the number of children have the same directional relationship with the propensity to migrate across all three racial and ethnic subgroups. Differences in the magnitudes of relationships were discussed in examining the results in Table 4. An incremental increase in the number of school-age children for White households, for example, has a larger negative impact on migration than for Black and Hispanic households. An addition of one school-age child in a Black household decreases the propensity to migrate by 0.084 percentage points, while it decreases the propensity to migrate for Hispanic households by 0.11 percentage points and 0.13 percentage points for White households. The magnitudes are similar for children under the age 5, and smaller than the impact of having school-age children.

[8]See Appendix B for the marginal effects without fixed effects and with only time fixed effects.

Table 4 Marginal Effects for Entire Sample (Selection=Migrate)

Variables	ALL		
Sex (Male=1)	0.02465***	0.02590***	0.02615***
	(0.001528)	(0.001533)	(0.001536)
Age	-0.04682***	-0.04686***	-0.04706***
	(0.0002811)	(0.0002812)	(0.0002817)
Age^2	0.0002493***	0.0002499***	0.0002519***
	(0.000002660)	(0.000002661)	(0.000002667)
Black	-0.01340***	-0.01254***	-0.02637***
	(0.002657)	(0.002658)	(0.002669)
Hispanic	-0.1046***	-0.1041***	-0.1250***
	(0.003086)	(0.003087)	(0.003098)
Marital Status (Married=1)	-0.2171***	-0.2178***	-0.2206***
	(0.001616)	(0.001617)	(0.001620)
# Of Children Age 5 and Up	-0.1236***	-0.1237***	-0.1223***
	(0.0009376)	(0.0009379)	(0.0009403)
# Of Children Under the Age of 5	-0.03728***	-0.03725***	-0.03492***
	(0.001737)	(0.001737)	(0.001740)
U.S. Born	-0.04440***	-0.04403***	-0.04044***
	(0.002871)	(0.002872)	(0.002876)
Less than High School	0.03425***	0.03398***	0.02661***
	(0.002879)	(0.002880)	(0.002885)
Some College	0.08782***	0.08790***	0.07815***
	(0.002180)	(0.002181)	(0.002190)
College	0.1259***	0.1257***	0.1195***
	(0.002076)	(0.002078)	(0.002085)
Advanced Degree	0.2461***	0.2461***	0.2408***
	(0.002508)	(0.002510)	(0.002518)
Labor Force Participant	-0.1789***	-0.1789***	-0.1749***
	(0.001986)	(0.001988)	(0.001992)
Unemployment Rate	-0.000454	0.002166***	0.002075***
	(0.0002582)	(0.0003622)	(0.0003669)
Total Population	-3.332e-08***	-3.338e-08***	-2.761e-08***
	(5.463e-10)	(5.438e-10)	(5.410e-10)
Per Capita Income	-0.000001029***	-0.000001045***	-5.390e-07***
	(6.468e-08)	(6.975e-08)	(7.218e-08)
% of Population that is Black	-0.3336***	-0.3357***	-0.3478***
	(0.01238)	(0.01243)	(0.01340)
% of Population that is White	-0.2204***	-0.2106***	-0.08917***
	(0.01159)	(0.01160)	(0.01208)
% of Population that is Hispanic	-0.09921***	-0.1106***	-0.09264***
	(0.01278)	(0.01278)	(0.01282)
Median Monthly Housing Cost	0.00004079***	0.00003875***	0.0001469***
	(0.000003278)	(0.000003372)	(0.000003802)
Per Capita Parks & Rec Expenditures	-4.158e-07***	-4.125e-07***	-4.683e-07***
	(1.253e-08)	(1.250e-08)	(1.214e-08)
Employment Growth	0.9929***	1.504***	1.234***
	(0.02571)	(0.03483)	(0.03480)
Time FE?	No	Yes	Yes
Region FE?	No	No	Yes

[a] Standard errors are in parentheses. * p < 0.1 ** p < 0.05 *** p < 0.01.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are of the residence of origin.

Table 5 Marginal Effects by Race & Ethnicity (Selection=Migrate) With Time & Region Fixed Effects

Variables	BLACK	WHITE	HISPANIC
Sex (Male=1)	0.06356***	0.01990***	0.04364***
	(0.005190)	(0.001697)	(0.005239)
Age	-0.03566***	-0.04891***	-0.03459***
	(0.001015)	(0.0003081)	(0.001099)
Age^2	0.0001473***	0.0002645***	0.0001711***
	(0.000009991)	(0.000002893)	(0.00001111)
Marital Status (Married=1)	-0.1935***	-0.2253***	-0.1798***
	(0.005924)	(0.001782)	(0.005335)
# of Children Age 5 and Up	-0.08365***	-0.1293***	-0.1057***
	(0.002613)	(0.001109)	(0.002559)
# of Children Under the Age of 5	-0.02058***	-0.03274***	-0.04463***
	(0.005708)	(0.001980)	(0.004863)
U.S. Born	-0.1143***	-0.05370***	0.04939***
	(0.008020)	(0.003894)	(0.005352)
Less than High School	0.03678***	0.05211***	-0.01478*
	(0.008037)	(0.003517)	(0.007107)
Some College	0.06347***	0.07653***	0.07201***
	(0.006557)	(0.002455)	(0.007427)
College	0.1028***	0.1151***	0.1359***
	(0.006855)	(0.002306)	(0.007465)
Advanced Degree	0.1686***	0.2393***	0.2620***
	(0.009055)	(0.002743)	(0.01002)
Labor Force Participation	-0.08984***	-0.1914***	-0.1277***
	(0.006187)	(0.002224)	(0.006855)
Unemployment Rate	0.007908***	0.002406***	0.004632***
	(0.001330)	(0.0004103)	(0.001403)
Total Population	-5.808e-08***	-2.662e-08***	-4.503e-08***
	(1.864e-09)	(6.685e-10)	(1.704e-09)
Per Capita Income	-9.584e-07***	-4.418e-07***	-0.000002073***
	(2.598e-07)	(7.949e-08)	(3.046e-07)
% of Population that is Black	-0.6035***	-0.2995***	0.2502***
	(0.05374)	(0.01452)	(0.05835)
% of Population that is White	-0.1482**	-0.07353***	0.4798***
	(0.05211)	(0.01285)	(0.04872)
% of Population that is Hispanic	-0.2363***	0.07560***	-0.4133***
	(0.05484)	(0.01421)	(0.04885)
Median Monthly Housing Cost	0.0003131***	0.00008687***	0.000002747
	(0.00001266)	(0.000004202)	(0.00001786)
Per Capita Parks & Rec Expenditures	-4.869e-07***	-4.471e-07***	-1.232e-07**
	(3.450e-08)	(1.507e-08)	(4.131e-08)
Employment Growth	1.947***	1.122***	1.487***
	(0.1195)	(0.03817)	(0.1522)
Time FE?	Yes	Yes	Yes
Region FE?	Yes	Yes	Yes

[a] Standard errors are in parentheses. * p < 0.1 ** p < 0.05 *** p < 0.01.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are of the residence of origin.

The educational attainment of the household head not only varies across race and ethnicity, but the relationship with migration behavior varies as well. A Black household head that has completed less than a high school degree is 0.37 percentage points more likely to migrate than is a household head that stopped at a high school degree. Hispanic households are 0.015 percentage points more likely while White households are 0.052 percentage points more likely. On the other end of the educational spectrum, a Black household with an advanced degree is 0.17 percentage points more likely to migrate, 0.24 percentage points more likely for White households, and 0.26 percentage points more likely for Hispanic households.

There is more variation in the results across race and ethnicity when examining location characteristics. A one percentage point increase in the unemployment rate of the original county of residence increases the propensity to migrate of a Black household head by 0.079 percentage points, 0.046 percentage points for Hispanic households, and 0.024 percentage for White households. One reason that economic shocks push migration decisions more for minority households is because minorities are disproportionately impacted during economic downturns. Monthly median housing costs also have relationships of different magnitudes. An increase in the monthly median housing costs of a particular state increases the likelihood of migration of Black households more than White and Hispanic household heads by more than 3 times. This could be consistent with Black households being more likely to experience racial discrimination in housing and forced migration due to gentrification.

As a measure of ethnogenesis, we include demographic population information. As the percentage of the population that is Black increases, Black households are significantly less likely to migrate than are other racial and ethnic households while Hispanic households are more likely to migrate. When the percentage of the population that is White increases, both Black and White households are less likely to migrate while Hispanic households migrate more. When the percentage of the population that is Hispanic increase, both Black and Hispanic households are less likely to migrate while White household migration is positively correlated.

Household and individual characteristics have similar directional relationships with the propensity to migrate (and be a part of the selection group) across race and ethnicity, with some variation in magnitudes, while location specific variables tend to have a greater impact on Black and Hispanic households. We see even more variation in the relationship between these variables and the

propensity to participate in interstate migration, as detailed in the next section.

5.2 Determinants of Interstate Migration

The selection equation is only one part of the internal migration process. Once a household migrates, the household is then envisioned to determine whether to migrate to a different state or stay within the same state. Table 6 provides marginal effects of choosing to participate in interstate migration (versus intrastate migration) for the entire sample.

For households that migrated within the last year, households with a Black or Hispanic household head are less likely to participate in interstate migration than are White households. Being married and having children also decreases the likelihood of migrating to a different state. Marriage and children can decrease residential mobility due to more members of a household having social networks. Even when a married couple and/or household with children migrate, they are more likely to stay closer to their current place of residence. Higher educational attainment positively correlates with interstate migration and labor force participation decreases the propensity to migrate to a different state.

Differences in location characteristics are calculated by finding the difference between the location characteristic of the new place of residence and the last place of residence. A positive difference in a variable means that the new place of residence has a higher value for that particular location characteristic. The marginal effect can be interpreted as a percentage point change in migrating to a different state with a one unit change in the difference of the location characteristic. For example, the coefficient of 0.29 for the location difference in employment growth tells us that a one percentage point increase in the difference in employment growth between the two locations, increases interstate migration by 0.29 percentage points. As another example, the coefficient of -0.0025 for the location difference in median monthly housing costs tells us that a one dollar increase in the difference in median monthly housing costs, decreases the likelihood of migrating to another state by 0.0025 percentage points. The negative sign indicates a lower propensity to migrate somewhere with higher housing costs and the positive sign for employment growth indicates a higher propensity to migrate to another state and location with higher employment growth.

Table 7 runs the same model on each racial and ethnic group with both time and region

Table 6 Marginal Effects for Entire Sample (Interstate)

Variables	ALL		
Sex (Male=1)	0.001486***	0.001536***	0.001535***
	(0.0001007)	(0.0001011)	(0.0001023)
Age	-0.001765***	-0.001769***	-0.001819***
	(0.00001975)	(0.00001976)	(0.00002000)
Age^2	0.000008533***	0.000008574***	0.000008917***
	(1.780e-07)	(1.780e-07)	(1.801e-07)
Black	-0.002335***	-0.002270***	-0.003677***
	(0.0001719)	(0.0001722)	(0.0001762)
Hispanic	-0.006407***	-0.006371***	-0.008453***
	(0.0001960)	(0.0001964)	(0.0002006)
Marital Status (Married=1)	-0.007547***	-0.007592***	-0.007899***
	(0.0001105)	(0.0001105)	(0.0001120)
# Of Children Age 5 and Up	-0.005335***	-0.005348***	-0.005347***
	(0.00006255)	(0.00006253)	(0.00006325)
# Of Children Under the Age of 5	-0.001738***	-0.001741***	-0.001602***
	(0.0001137)	(0.0001138)	(0.0001152)
U.S. Born	-0.002848***	-0.002835***	-0.002575***
	(0.0001802)	(0.0001802)	(0.0001818)
Less than High School	-0.0003481	-0.0003557	-0.0007454***
	(0.0002037)	(0.0002039)	(0.0002064)
Some College	0.005280***	0.005332***	0.004680***
	(0.0001495)	(0.0001497)	(0.0001516)
College	0.008957***	0.008981***	0.008563***
	(0.0001405)	(0.0001407)	(0.0001424)
Advanced Degree	0.01660***	0.01663***	0.01628***
	(0.0001652)	(0.0001655)	(0.0001675)
Labor Force Participation	-0.01116***	-0.01115***	-0.01095***
	(0.0001312)	(0.0001313)	(0.0001325)
Employment Growth	0.02669***	0.03716***	0.02935***
	(0.001812)	(0.001861)	(0.001834)
Total Population	-8.806e-10***	-8.833e-10***	-8.039e-10***
	(1.874e-11)	(1.876e-11)	(1.812e-11)
% of Population that is Black	-0.01605***	-0.01604***	-0.01572***
	(0.0006809)	(0.0006795)	(0.0006707)
% of Population that is Black White	-0.01328***	-0.01309***	-0.01009***
	(0.0006022)	(0.0006013)	(0.0005880)
% of Population that is Hispanic	-0.006620***	-0.006818***	-0.007376***
	(0.0006682)	(0.0006672)	(0.0006433)
Per Capita Income	-0.000006983***	-0.000006907***	-0.000006062***
	(2.499e-07)	(2.512e-07)	(2.570e-07)
Median Monthly Housing Cost	-0.002651***	-0.002657***	-0.002547***
	(0.00002518)	(0.00002516)	(0.00002508)
Unemployment Rate	-0.00008936***	-0.0000358	-0.0000438
	(0.00002209)	(0.00002277)	(0.00002245)
Per Capita Parks & Rec Expenditures	-0.1597***	-0.1587***	-0.1658***
	(0.004609)	(0.004603)	(0.004496)
Time FE?	No	Yes	Yes
Region FE?	No	No	Yes

[a] Standard errors are in parentheses. * p < 0.1 ** p < 0.05 *** p < 0.01.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are the differences between the new place of residence and the original place of residence.

fixed effects.[9] Unlike the relationships between individual and household characteristics on the propensity to migrate, the relationships between the same characteristics on interstate migration vary for the different households. Age shares a negative relationship with the propensity to migrate to a different state for Black and White households while the opposite is true for Hispanic households. Age, including the quadratic, differs for Hispanic households.

Marital status is negatively correlated with interstate migration of Black and White household while sharing a positive relationship for Hispanic households. Like for the entire sample, having school-aged children decreases the propensity for Black and White households, however this family structure increases the propensity for Hispanic households. Having younger children does not share a statistically significant relationship with interstate migration for Hispanic households. Like in the selection model, the magnitudes of the impacts are different across race and ethnicity. In addition to the different directional relationship between age and interstate migration across race and ethnicity, the magnitudes of coefficients are different. For both Black and White households, males are more likely to participate in interstate migration. Black male households are 0.003 percentage points more likely while White households are more than half that, at 0.0014 percentage points. We continue to see differences in magnitudes for the number of children the household has, whether the household head was born in the U.S., median monthly housing costs, the unemployment rate, and many of the educational attainment variables.

6 Conclusion

When controlling for individual and location characteristics down to the county level, the results presented in this paper align with the findings that minorities are less likely to migrate. Not only does this paper provide an overview of determinants of migration but also illustrates how these types of independent variables relate to the propensity to migrate differently across race and ethnicity. The signs of the coefficients for household and individual characteristics are relatively consistent across race and ethnicity, aligning with previous migration literature. We find more variation in the directions and magnitudes of coefficients for location characteristics and for migration behaviors across the three racial and ethnic groups examined such as differences

[9] See Appendix B for the marginal effects without fixed effects and with only time fixed effects.

Table 7 Marginal Effects by Race & Ethnicity (Interstate) With Time & Region Fixed Effects

Variables	BLACK	WHITE	HISPANIC
Sex (Male=1)	0.003025***	0.001372***	-0.0008978
	(0.0003213)	(0.0001194)	(0.004100)
Age	-0.001253***	-0.001996***	0.006453***
	(0.00006574)	(0.00002330)	(0.0008685)
Age^2	0.000004175***	0.000009906***	-0.00004327***
	(6.355e-07)	(2.074e-07)	(0.000009089)
Marital Status (Married=1)	-0.007024***	-0.008459***	0.03600***
	(0.0003743)	(0.0001305)	(0.004241)
# of Children Age 5 and Up	-0.003230***	-0.006029***	0.01612***
	(0.0001600)	(0.00007882)	(0.001994)
# of Children Under the Age of 5	-0.0008927*	-0.001618***	0.001255
	(0.0003485)	(0.0001385)	(0.003639)
U.S. Born	-0.006895***	-0.002950***	-0.03682***
	(0.0004643)	(0.0002583)	(0.004555)
Less than High School	-0.0002823	-0.0003813	-0.0009455
	(0.0005282)	(0.0002725)	(0.005680)
Some College	0.003538***	0.005060***	-0.007244
	(0.0004165)	(0.0001799)	(0.005804)
College	0.006154***	0.009183***	0.01519*
	(0.0004265)	(0.0001670)	(0.006327)
Advanced Degree	0.01142***	0.01746***	0.06044***
	(0.0005441)	(0.0001941)	(0.01072)
Labor Force Participation	-0.005282***	-0.01262***	-0.01818**
	(0.0003842)	(0.0001564)	(0.006126)
Employment Growth	0.03230***	0.03096***	-0.1273
	(0.005976)	(0.002123)	(0.08712)
Total Population	-1.450e-09***	-8.156e-10***	6.851e-09***
	(6.119e-11)	(2.379e-11)	(8.409e-10)
% of Population that is Black	-0.009408***	-0.01720***	-0.1343***
	(0.002115)	(0.0007726)	(0.03157)
% of Population that is White	-0.004749*	-0.01139***	-0.06616*
	(0.002041)	(0.0006637)	(0.02577)
% of Population that is Hispanic	-0.0004297	-0.004371***	-0.01373
	(0.002189)	(0.0007614)	(0.02611)
Par Capita Income	-0.000007682***	-0.000005919***	-0.00007600***
	(8.102e-07)	(3.058e-07)	(0.00001413)
Median Monthly Housing Cost	-0.002291***	-0.002707***	-0.02257***
	(0.00007688)	(0.00002974)	(0.002095)
Unemployment Rate	0.0001029	-0.00002257	-0.005016***
	(0.00007486)	(0.00002668)	(0.0009510)
Per Capita Parks & Rec Expenditures	-0.2035***	-0.1488***	-1.838***
	(0.01251)	(0.005346)	(0.2000)
Time FE?	Yes	Yes	Yes
Region FE?	Yes	Yes	Yes

[a] Standard errors are in parentheses. * p < 0.1 ** p < 0.05 *** p < 0.01.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are the differences between the new place of residence and the original place of residence.

in the coefficients for unemployment rate, per capita income, and racial and ethnic population composition.

Unlike the major results from the selection stage of the model, there are notable directional differences in the relationship between household characteristics and the propensity to participate in internal migration. Hispanic households consistently have opposite directional relationships with Black and White households such as a positive relationship between age and interstate migration, where Black and White households have a negative relationship (consistent with migration literature) and the number of children within the household. Many of these differences for Hispanic households are likely a product of the large differences in family structure and culture.

This paper examines not only the determinants of migration and interstate migration but how the signs and magnitudes of the coefficients vary across race and ethnicity. The ability to use restricted-use ACS data allowed for a more narrow definition of migration, separating residential movement within counties and internal migration. It builds a framework for understanding the differences in residential mobility across race and ethnicity.

7 Conclusion

This work contributes to existing literature to provide a deeper understanding of the different levels of residential movement and migration. With racial and ethnic disparities present both historically and in today's society, residential movement and migration is no exception. Without restricted-use ACS data, it would not be possible to analyze within county moves and how the determinants of movers differ for internal migrants at a national level across subpopulations. As hypothesized, differences in not only directional relationships between types of movement and migration but also in the magnitudes in which the variables impact movement and migration are identified. While the number of children decreases the propensity to participate in intra- or interstate migration relative to moving within county, the size of the impact on the propensity differs across race and ethnicity in the direction of having a larger negative impact on White households than on Black or Hispanic households for both intra- and interstate migration relative to the reference group. There are also differences in county-level unemployment rates, employment growth, and population composition.

This work contributes to the existing literature on internal migration by examining the role that both economic and noneconomic location and household characteristics play in the migratory behaviors of U.S. households. It melds together sociology and economic theories such as ethnogensis and how the racial composition of a location's population can influence migration decisions across Black, White, and Hispanic households. Private-use data allows for the examination of smaller geographic residential movement, in particular within county moves that is likely capturing residential displacement. By separating within county moves and migration, this work is better able to capture different types of residential movement across race and ethnicity.

Household and individual characteristics share similar relationships with the propensity to migrate across all three racial and ethnic group, however the magnitudes of the coefficients varied. There was more variation in the impact of location characteristics on migratory behaviors for variables like the unemployment rate, per capita income, and the racial and ethnic population composition of the final destination. When a household does migrate, the impact on whether the household participates in interstate versus intrastate migration greatly differs across race and ethnicity. Hispanic households consistently experience opposite directional relationships between the explanatory variables and the propensity to migrate to a different state than do Black and White households. The variables that stand out are age and the number of children, likely driven by the difference in culture and familial structure.

Due to data availability, median monthly housing costs were collected at the state level. This could make interpreting the coefficients difficult for the groups that moved within county or participated in intrastate migration as the difference in median monthly housing costs for these two groups would be zero. Additionally, the population composition variables provided some unexpected results such as a negative sign for the differences in population composition for White households. A further dive into these variables and alternative measures to proxy for ethnogenesis could further contribute to the literature.

Future modifications could include updating data as it becomes available and including more county-level location characteristics such as average January temperatures that have been shown to impact migration behavior and other variables that measure location-specific amenities that could increase the benefits (or costs) of moving or migrating to that location such as crime rates or other location amenity variables.

Finally, restricted-use data of the type used in this paper could be used for further locational analysis down to the Census-tract level if place characteristics are calculated at that level. Future research in this direction therefore could reach understandings of the role of individual neighborhoods as smaller geographies than the counties explored here.

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Appendices

A Theoretical Model

Drawing from conceptual frameworks in the reviewed literature, we outline a theoretical model to explain a household's decision to migrate. While this model can be used to explain migration in a broad sense, the estimation presented in this paper focuses on internal migration decisions of a household within the United States.

A.1 Benefit-Cost Analysis in Self-Selection into Migration

In theory, when deciding where to migrate, a potential migrant weighs the costs and benefits of the move. If the real value of expected benefits exceed the expected costs, ceteris paribus, then the individual migrates. As outlined in Greenwood (1975), the present value of investment of migration from place i to place j is as follows:

$$PV_{ij} = \sum_{t=1}^{n} \frac{E_{jt} - E_{it}}{(1+r)^t} - \sum_{t=1}^{n} \frac{C_{jt} - C_{it}}{(1+r)^t} \tag{3}$$

In time period t, $E_{jt} - E_{it}$ represents the difference between the present discounted value of earnings in location j and i and r represents the discount rate. $C_{jt} - C_{it}$ represents the difference in residence costs between location j and i. Given the potential earning streams of the individual and the costs associated with residing in each location, the potential migrant residing in location i migrates to location j if the present value of investment to migration is positive. This can be envisioned in either a real or an expected value sense. While current and future income are important in migration decisions, behavioral and location characteristics can influence not only migration but the location of migration.

A.2 Utility Maximization of the Household

Previous studies on the determinants of migration have commonly explained migration decision within the context of utility maximization (Greenwood 1997). A utility maximizing framework allows for households to value things differently which is important when analyzing the differences

in internal migration between groups. There are two questions a potential migrant needs to answer. The first question is whether to migrate and the second is where to migrate. While the migration literature addresses these two questions, there are disagreements regarding whether these questions are answered simultaneously or in some particular order. We model these questions as being dependent on one another. If the individual decides whether to migrate and then where to migrate, they will consider both household characteristics and location characteristics in order to maximize utility. Migration occurs if an individual is able to increase their utility by migrating. An increase in utility can be a result of "better employment opportunities, higher wages, a preferred bundle of amenities, and many other factors" (Greenwood 1997).

Like other models of migration using utility maximization, Graves and Linneman (1997) assume that the household's goal is to maximize lifetime utility dependent on the household's lifetime income and leisure. Following the notation of Graves and Linneman (1977), the household receives utility every year from year 1 to year n such that:

$$U = U(u_1, \ldots, u_i, \ldots, u_n) \tag{4}$$

Where u represents the annual utility of the household for each year from 1 to n and U represents lifetime utility. Based on the categories of goods outlined in Tolley (1974), each household can consume traded goods, which are not specific to the location, and non-traded goods, which are location-specific. Examples of non-traded goods can include location specific public goods, availability of schools for children, and entertainment. Each year's utility of household m is a function of traded goods consumed, X_m; non-traded goods consumed in a particular location l, Z_l; and leisure, L_m. Household m maximizes the following lifetime utility, U:

$$U = U(X_m, Z_l, L_m) \tag{5}$$

The household's lifetime utility is subject to a time constraint. Total lifetime available time,

T_m, depends on lifetime labor supply, S_m, and lifetime leisure, L_m:

$$T_m = S_{HH} + L_{HH} \tag{6}$$

$$T_m = S_{HH} + S_{SP} + L_{HH} + L_{SP} \tag{7}$$

Where equation (6) represents the household's total available time for a single household head. The household's total lifetime labor supply, S_m, is equal to the lifetime labor supply of a single household head, S_{HH}, in household m.[10] Like labor supply, household m's leisure time, L_m, is equal to the lifetime leisure of a single household, L_{HH}. Equation (7) represents the household's total available time for a married household head. Total labor supply, S_m, is equal to the labor supply of the household head, S_{HH}, and the labor supply of the spouse, S_{SP}, in household m. The household's lifetime consumption of leisure, L_m, is the sum of lifetime leisure of the household head, L_{HH}, and the lifetime leisure of the spouse, L_{SP}, in household m.

The household's utility is subjected to an income constraint:

$$Y_m = W_{HH}S_{HH} + V_{HH} \tag{8}$$

$$Y_m = W_{HH}S_{HH} + W_{SP}S_{SP} + V_{HH} + V_{SP} \tag{9}$$

Where equation (8) represents the total lifetime household income of a single household head and equation (9) outlines the household's lifetime income of a dual earning household. Total income of an unmarried head is a function of their labor supply, S_{HH}, their wages, W_{HH} and non-labor income, V_{HH}. Total income of a dual earning household is a function of the labor supply of both the household head the spouse, S_{SP}, the labor wages of the household head and the spouse, W_{SP}, and the non-labor income of the household head and the spouse, V_{SP}. It is important to note that when the household is maximizing their utility and deciding on whether or not to migrate, wages will be a function of the location. When estimating lifetime utility of staying in location i, wages of location i are used in calculating lifetime wages. The potential rate of wage growth is also a function of the location's projected economic growth. When calculating wages lifetime utility of location j, wages of that location are used. The household's lifetime utility maximization problem

[10]The m subscript is suppressed from S_{HH}, L_{HH}, S_{SP}, L_{SP}, W_{HH}, W_{SP}, V_{HH}, and V_{SP} to simplify notation.

can be rewritten and simplified as:

$$maxU = U(X_m, Z_l, L_m) \tag{10}$$

Subject to:

$$W_{HH}S_{HH} + W_{SP}S_{SP} + V_{HH} + V_{SP} = P_xX_m + P_zZ_l + W_{HH}L_{HH} + W_{SP}L_{SP} \tag{11}$$

$$T_m = S_{HH} + S_{SP} + L_{HH} + L_{SP} \tag{12}$$

$$W_{SP}, S_{SP}, V_{SP}, L_{SP} = 0 \textit{ if household head is single} \tag{13}$$

$$X_m, Z_l, L_m \geq 0 \tag{14}$$

Using the above maximization problem, the Lagrangian takes the following form:

$$\ell(X_m, Z_m, L_m, \lambda) = U(X_m, Z_m, L_m) - \lambda(W_{HH}S_{HH} + W_{SP}S_{SP} + V_{HH} + V_{SP} - P_xX_m - P_zX_m - P_zZ_m - W_{HH}L_{HH} - W_{SP}L_{SP}) \tag{15}$$

A.3 Migration Decision of the Household

The household's decision to migrate and location of migration depends on the particular household's utility maximization problem. Recall that a household can consume traded goods in any location but only consume non-traded goods in a particular location. In order to simplify this problem, suppose the household has the choice of staying in location i or migrating to location j. The household will migrate to location j if:

$$maxU(X_m, Z_j, L_m) > maxU(X_m, Z_i, L_m)$$

The household will stay in location i if:

$$maxU(X_m, Z_j, L_m) < maxU(X_m, Z_i, L_m)$$

Each household behaves differently and their utility function is determined by household

characteristics. Each household's utility is a function of the consumption of traded goods, the consumption of location specific goods, additional location specific characteristics that may affect income, and the need for leisure. Finally, the importance of these characteristics may vary by racial and ethnic groups as tested empirically in this paper.

B Heckman Results With Varying Fixed Effects

The following tables provide additional Heckman corrected results. These results, for comparison, are presented without fixed effects and with only time fixed effects for the entire sample and across the three racial and ethnic groups examined throughout this work.

Table 8 Marginal Effects by Race & Ethnicity (Selection=Migrate) Without Fixed Effects

Variables	BLACK	WHITE	HISPANIC
Sex (Male=1)	0.06381***	0.01769***	0.04430***
	(0.005174)	(0.001687)	(0.005226)
Age	-0.03594***	-0.04849***	-0.03480***
	(0.001014)	(0.0003073)	(0.001097)
Age^2	0.0001496***	0.0002599***	0.0001730***
	(0.000009978)	(0.000002885)	(0.00001109)
Marital Status (Married=1)	-0.1857***	-0.2229***	-0.1753***
	(0.005903)	(0.001777)	(0.005320)
# Of Children Age 5 and Up	-0.08477***	-0.1310***	-0.1064***
	(0.002607)	(0.001105)	(0.002556)
# Of Children Under the Age of 5	-0.02245***	-0.03552***	-0.04502***
	(0.005703)	(0.001975)	(0.004858)
U.S. Born	-0.1048***	-0.06202***	0.04533***
	(0.007982)	(0.003882)	(0.005320)
Less than High School	0.03934***	0.06092***	-0.01188
	(0.008025)	(0.003505)	(0.007093)
Some College	0.06510***	0.08935***	0.07456***
	(0.006531)	(0.002442)	(0.007414)
College	0.1056***	0.1237***	0.1387***
	(0.006835)	(0.002294)	(0.007453)
Advanced Degree	0.1724***	0.2470***	0.2621***
	(0.009031)	(0.002730)	(0.01001)
Labor Force Participation	-0.08549***	-0.1979***	-0.1262***
	(0.006170)	(0.002217)	(0.006842)
Unemployment Rate	4.77E-05	-0.00019	-0.002107*
	(0.0008572)	(0.0002905)	(0.001016)
Total Population	-6.392e-08***	-3.369e-08***	-4.719e-08***
	(1.727e-09)	(6.847e-10)	(1.658e-09)
Per Capita Income	-0.000002627***	-7.818e-07***	-0.000003376***
	(2.283e-07)	(7.146e-08)	(2.756e-07)
% of Population that is Black	-0.7685***	-0.2551***	0.1520**
	(0.04651)	(0.01364)	(0.04882)
% of Population that is White	-0.4609***	-0.1653***	0.2807***
	(0.04666)	(0.01259)	(0.04484)
% of Population that is Hispanic	-0.3628***	0.1089***	-0.4794***
	(0.05039)	(0.01460)	(0.04498)
Median Monthly Housing Cost	0.0001909***	-0.00001360***	-1E-05
	(0.00001089)	(0.000003644)	(0.00001366)
Per Capita Parks & Rec Expenditures	-4.142e-07***	-3.926e-07***	-8.409e-08*
	(3.408e-08)	(1.557e-08)	(4.072e-08)
Employment Growth	1.251***	0.9431***	1.298***
	(0.08772)	(0.02867)	(0.1089)
Time FE?	No	No	No
Region FE?	No	No	No

[a] Standard errors are in parentheses. * p < 0.1 ** p < 0.05 *** p < 0.01.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are of the residence of origin.

Table 9 Marginal Effects by Race & Ethnicity (Selection=Migrate) With Time Fixed Effects

Variables	BLACK	WHITE	HISPANIC
Sex (Male=1)	0.06393***	0.01966***	0.04374***
	(0.005182)	(0.001693)	(0.005233)
Age	-0.03588***	-0.04858***	-0.03475***
	(0.001014)	(0.0003075)	(0.001098)
Age^2	0.0001493***	0.0002611***	0.0001727***
	(0.000009977)	(0.000002887)	(0.00001110)
Marital Status (Married=1)	-0.1880***	-0.2235***	-0.1765***
	(0.005910)	(0.001778)	(0.005324)
# Of Children Age 5 and Up	-0.08486***	-0.1310***	-0.1064***
	(0.002610)	(0.001106)	(0.002556)
# Of Children Under the Age of 5	-0.02227***	-0.03548***	-0.04526***
	(0.005705)	(0.001976)	(0.004859)
U.S. Born	-0.1057***	-0.06162***	0.04634***
	(0.007983)	(0.003884)	(0.005323)
Less than High School	0.03857***	0.06051***	-0.01228
	(0.008028)	(0.003507)	(0.007095)
Some College	0.06553***	0.08946***	0.07457***
	(0.006538)	(0.002443)	(0.007417)
College	0.1054***	0.1236***	0.1385***
	(0.006841)	(0.002296)	(0.007456)
Advanced Degree	0.1718***	0.2471***	0.2620***
	(0.009041)	(0.002733)	(0.01001)
Labor Force Participation	-0.08748***	-0.1973***	-0.1268***
	(0.006176)	(0.002218)	(0.006846)
Unemployment Rate	0.003168*	0.003218	0.001956
	(0.001259)	(0.0004062)	(0.001363)
Total Population	-6.396e-08***	-3.375e-08***	-4.747e-08***
	(1.772e-09)	(6.801e-10)	(1.661e-09)
Per Capita Income	-0.000002524***	-7.676e-07***	-0.000002985***
	(2.419e-07)	(7.733e-08)	(2.951e-07)
% of Population that is Black	-0.7341***	-0.2652***	0.1481**
	(0.04685)	(0.01371)	(0.04908)
% of Population that is White	-0.4234***	-0.1601***	0.2905***
	(0.04686)	(0.01261)	(0.04497)
% of Population that is Hispanic	-0.3625***	0.09465***	-0.4862***
	(0.05056)	(0.01461)	(0.04515)
Median Monthly Housing Cost	0.0001960***	-0.00001826***	-2.3E-05
	(0.00001096)	(0.000003740)	(0.00001452)
Per Capita Parks & Rec Expenditures	-4.067e-07***	-3.900e-07***	-8.447e-08*
	(3.440e-08)	(1.548e-08)	(4.082e-08)
Employment Growth	2.247***	1.339***	1.963***
	(0.1166)	(0.03836)	(0.1498)
Time FE?	Yes	Yes	Yes
Region FE?	No	No	No

[a] Standard errors are in parentheses. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are of the residence of origin.

Table 10 Marginal Effects by Race & Ethnicity (Interstate) Without Fixed Effects

Variables	BLACK	WHITE	HISPANIC
Sex (Male=1)	0.003135***	0.001287***	-0.00169
	(0.0003206)	(0.0001168)	(0.003439)
Age	-0.001253***	-0.001904***	0.006118***
	(0.00006546)	(0.00002286)	(0.0007462)
Age^2	0.000004208***	0.000009232***	-0.00003980***
	(6.332e-07)	(2.037e-07)	(0.000007731)
Marital Status (Married=1)	-0.006732***	-0.008040***	0.03250***
	(0.0003721)	(0.0001280)	(0.003641)
# of Children Age 5 and Up	-0.003201***	-0.006017***	0.01590***
	(0.0001588)	(0.00007754)	(0.001691)
# of Children Under the Age of 5	-0.0009151**	-0.001791***	0.002459
	(0.0003493)	(0.0001358)	(0.003055)
U.S. Born	-0.006491***	-0.003661***	-0.03152***
	(0.0004616)	(0.0002538)	(0.003937)
Less than High School	-0.00021	9.98E-05	-0.00067
	(0.0005261)	(0.0002665)	(0.004776)
Some College	0.003523***	0.006032***	-0.0083
	(0.0004128)	(0.0001766)	(0.004884)
College	0.006239***	0.009849***	0.00897
	(0.0004247)	(0.0001639)	(0.005160)
Advanced Degree	0.01142***	0.01802***	0.04617***
	(0.0005400)	(0.0001906)	(0.008628)
Labor Force Participation	-0.005243***	-0.01294***	-0.01248*
	(0.0003834)	(0.0001542)	(0.004997)
Employment Growth	0.01654**	0.03081***	-0.1398
	(0.005775)	(0.002085)	(0.07275)
Total Population	-1.521e-09***	-9.243e-10***	6.270e-09***
	(5.976e-11)	(2.523e-11)	(7.311e-10)
% of Population that is Black	-0.009394***	-0.01756***	-0.1081***
	(0.002092)	(0.0007934)	(0.02632)
% of Population that is White	-0.006666**	-0.01451***	-0.03796
	(0.002029)	(0.0006896)	(0.02143)
% of Population that is Hispanic	0.000669	-0.002526**	0.000483
	(0.002181)	(0.0008054)	(0.02169)
Par Capita Income	-0.000009304***	-0.000006808***	-0.00005645***
	(7.702e-07)	(2.971e-07)	(0.00001155)
Median Monthly Housing Cost	-0.002316***	-0.002822***	-0.01834***
	(0.00007584)	(0.00002983)	(0.001747)
Unemployment Rate	-9.4E-06	-0.00006463*	-0.003874***
	(0.00007073)	(0.00002617)	(0.0008013)
Per Capita Parks & Rec Expenditures	-0.2081***	-0.1403***	-1.470***
	(0.01185)	(0.005468)	(0.1641)
Time FE?	No	No	No
Region FE?	No	No	No

[a] Standard errors are in parentheses. * p < 0.1 ** p < 0.05 *** p < 0.01.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are the differences between the new place of residence and the original place of residence.

Table 11 Marginal Effects by Race & Ethnicity (Interstate) With Time Fixed Effects

Variables	BLACK	WHITE	HISPANIC
Sex (Male=1)	0.003100***	0.001382***	-0.00184
	(0.0003217)	(0.0001174)	(0.003538)
Age	-0.001256***	-0.001909***	0.006180***
	(0.00006463)	(0.00002288)	(0.0007643)
Age^2	0.000004259***	0.000009280***	-0.00004038***
	(6.229e-07)	(2.038e-07)	(0.000007933)
Marital Status (Married=1)	-0.006815***	-0.008083***	0.03287***
	(0.0003728)	(0.0001281)	(0.003726)
# of Children Age 5 and Up	-0.003223***	-0.006031***	0.01597***
	(0.0001599)	(0.00007758)	(0.001733)
# of Children Under the Age of 5	-0.0008788*	-0.001796***	0.002212
	(0.0003489)	(0.0001360)	(0.003143)
U.S. Born	-0.006502***	-0.003676***	-0.03212***
	(0.0004595)	(0.0002538)	(0.004029)
Less than High School	-0.0002	(0.0002669)	-0.00089
	(0.0005252)		(0.004912)
Some College	0.003724***	0.006070***	-0.00832
	(0.0004140)	(0.0001769)	(0.005022)
College	0.006342***	0.009853***	0.01014
	(0.0004246)	(0.0001642)	(0.005344)
Advanced Degree	0.01158***	0.01804***	0.04857***
	(0.0005418)	(0.0001910)	(0.008959)
Labor Force Participation	-0.005226***	-0.01293***	-0.01337**
	(0.0003835)	(0.0001543)	(0.005159)
Employment Growth	0.03494***	0.03951***	-0.1883*
	(0.005875)	(0.002153)	(0.07527)
Total Population	-1.509e-09***	-9.270e-10***	6.434e-09***
	(5.953e-11)	(2.525e-11)	(7.506e-10)
% of Population that is Black	-0.008940***	-0.01764***	-0.1108***
	(0.002121)	(0.0007924)	(0.02708)
% of Population that is White	-0.006167**	-0.01437***	-0.04003
	(0.002060)	(0.0006887)	(0.02205)
% of Population that is Hispanic	0.00064	-0.002768***	-0.00029
	(0.002214)	(0.0008050)	(0.02233)
Par Capita Income	-0.000009141***	-0.000006702***	-0.00006003***
	(7.694e-07)	(2.998e-07)	(0.00001191)
Median Monthly Housing Cost	-0.002306***	-0.002832***	-0.01881***
	(0.00007737)	(0.00002987)	(0.001796)
Unemployment Rate	3.2E-05	9.27E-06	-0.004256***
	(0.00007372)	(0.00002712)	(0.0008272)
Per Capita Parks & Rec Expenditures	-0.2065***	-0.1390***	-1.5190***
	(0.01219)	(0.005459)	(0.1693)
Time FE?	Yes	Yes	Yes
Region FE?	No	No	No

[a] Standard errors are in parentheses. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$.
[b] All output, including observation numbers, have been rounded in order to adhere to privacy requirements when using restricted-use data.
[c] Location characteristics are the differences between the new place of residence and the original place of residence.